

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Reusable plug, Part Number 4918875
- Gasket scraper or cleaning pad, Part Number 3823258
- Mounting tool, Part Number ST-1325, and indicator, Part Number 3376050.

Additional Service Items

- Hoist and lifting sling

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

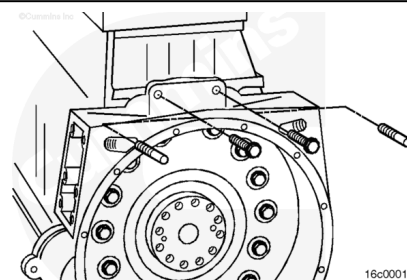
Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause personal injury.

- Disconnect the batteries and air starter to prevent accidental engine starting. See the equipment manufacturer service information.
- Remove the transmission, clutch, and all related components. See the equipment manufacturer service information.
- Remove the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/377/377-016-005.html>)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-037.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)
- Remove the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-024.html>)
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html>)
- Remove the rear engine mounts from the flywheel housing. Refer to Procedure 016-010 in Section 16. (</qs3/pubsys2/xml/en/procedures/00/00-016-010.html>)

Remove

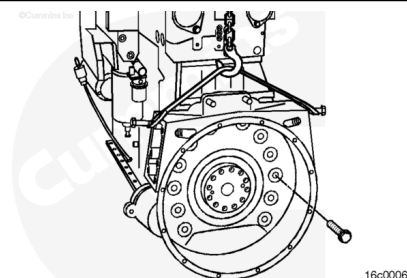
Remove the top exterior capscrews from the flywheel housing.

Install guide studs.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



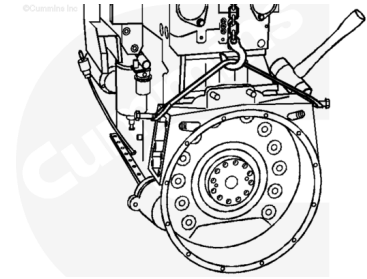
Use a hoist and lifting sling.

Install the lifting sling around both sides of the upper housing.

Adjust the hoist until there is tension in the lifting sling.

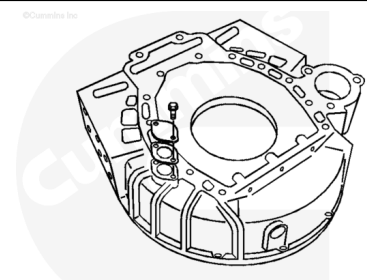
Remove the remaining capscrews.

Use a mallet to tap the flywheel housing off the locating dowels.



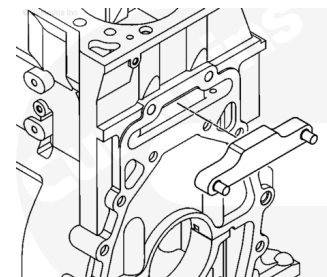
Remove the capscrews, cover plate, and gasket.

Discard the gasket.



⚠ CAUTION ⚠

Immediately upon removal of the flywheel housing assembly, a plug must be inserted into the cylinder block oil passage drillings. Failure to insert the oil passage plug can result in a bearing failure, crankshaft failure, or both.



The reusable plug, Part Number 4918875, is necessary to prevent debris from entering the lubricating system during the repair.

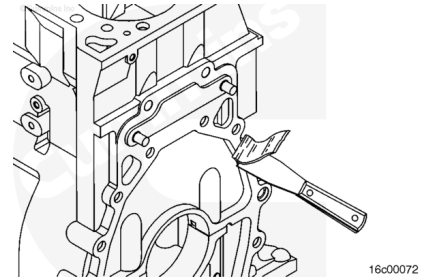
Plug the cylinder block oil passage with plug, Part Number 4918875.

Firmly push the plug into the oil passage to prevent debris from entering the lubricating system when cleaning the gasket surface.

Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not allow dirt or foreign material to enter oil passages in the cylinder block when cleaning the gasket sealing surfaces. Main or connecting rod bearing failures can be caused if debris is introduced into the lubrication system. Therefore, use of power tools combined with abrasive pads to clean gasket surfaces is not recommended.

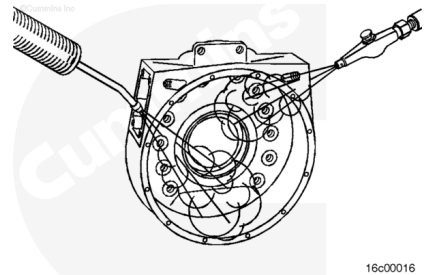


Clean the gasket surfaces by hand with a gasket scraper or cleaning pad, Part Number 3823258.

If the residual gasket material can **not** be felt with a finger, the surface is ready to accept the new gasket.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

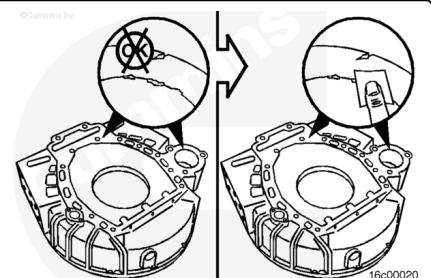
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use steam or solvent to clean the flywheel housing.

Dry with compressed air.

Inspect all surfaces for nicks, burrs, or cracks.

Use a fine crocus cloth to remove small nicks and burrs.



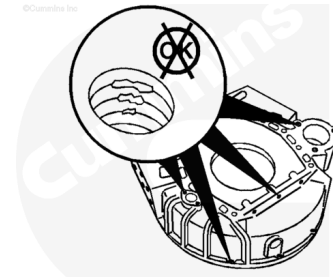
Inspect the rear face of the gear housing and flywheel housing mounting surfaces for cleanliness and raised nicks, burrs or cracks.

Replace the flywheel housing if cracks are present.

Thoroughly clean the flywheel housing and gear housing mating surfaces. These surfaces **must** be clean and free of oil and debris.

Inspect for damaged threads, commonly caused by cross-threaded capscrews or installing an incorrect capscrew.

Repair or replace the flywheel housing if the capscrew holes are damaged.

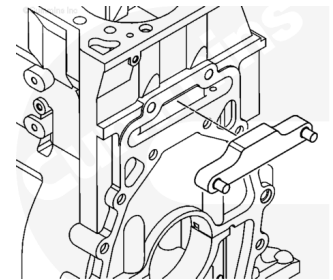


16c00032

Install

Remove the protective plug from the cylinder block oil passages. Make sure debris does **not** enter the lubricating system.

Use a clean shop towel to carefully wipe out the oil passage. Inspect the passage to make sure no debris is left.

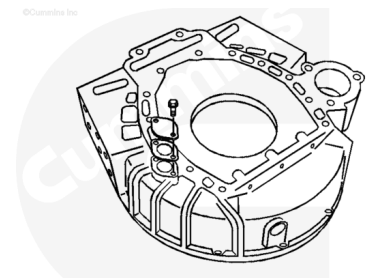


16c00071

Install the cover plate and capscrews.

Tighten the capscrews.

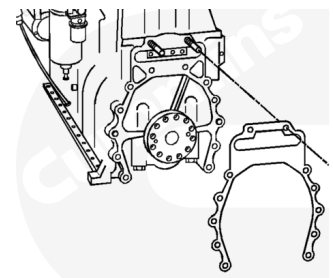
Torque Value: 27 n•m [239 in-lb]



16c00035

Install guide studs into the cylinder block to help support and align the housing during installation.

Position the flywheel housing gasket to the rear of the engine block using the guide pins.



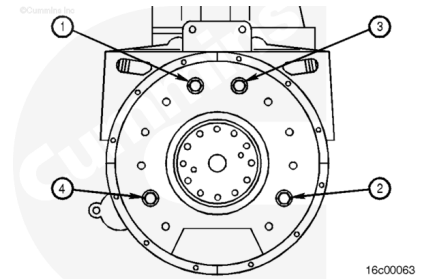
16c00062

Install the flywheel housing over the guide studs.

Lubricate the threads and underneath the heads of the capscrews with clean engine lubricating oil.

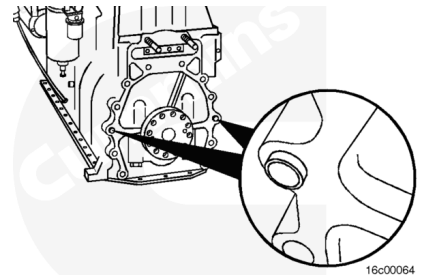
Install capscrews 1, 2, 3, and 4. Tighten the capscrews in the sequence illustrated.

Torque Value: 7 n·m [62 in-lb]



The flywheel housing is located on the cylinder block from the factory by two ring dowels. If a flywheel housing is removed, it can be installed with the ring dowels in place, but alignment **must** be checked with the following alignment procedure.

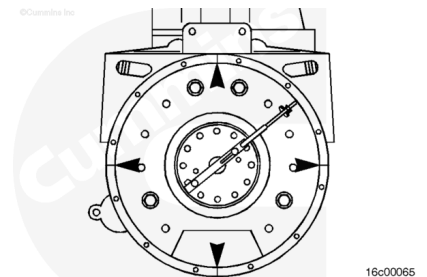
If the alignment does **not** meet specification, the ring dowels **must** be removed and the housing **must** be realigned without the ring dowels.



⚠ CAUTION ⚠

The flywheel housing bore and surface must be in alignment with the crankshaft to prevent possible damage to the engine, the clutch, or the transmission.

Measure the flywheel housing alignment.



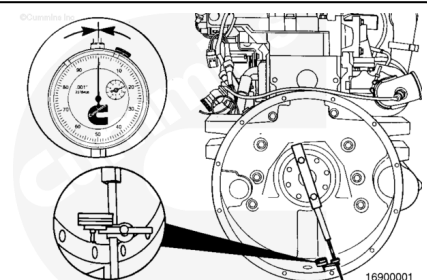
Measure

Bore Alignment

Attach a dial indicator to the crankshaft. Use mounting tool, Part Number ST-1325, and indicator, Part Number 3376050.

The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid and so it does **not** sag. If the bar sags or the indicator slips, the readings obtained will be inaccurate.

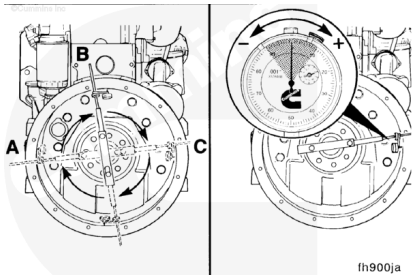
Position the indicator in the 6:00 o'clock position and set the gauge to zero.



Slowly rotate the crankshaft. Record the readings obtained at the 9-o'clock, 12-o'clock, and 3-o'clock positions as (a), (b), and (c) in the concentricity work sheet. Recheck zero at the 6-

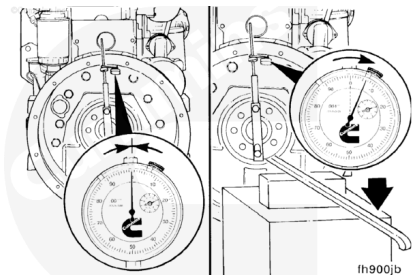
o'clock position.

The values for (a), (b), and (c) can be positive or negative. Use the accompanying illustration to determine the correct sign when recording these values.



⚠ CAUTION ⚠

Do not force the crankshaft beyond the point where the bearing clearance has been removed. Do not pry against the flywheel housing. These actions can cause false bearing clearance reading.



Rotate the crankshaft until the dial indicator is at the 12-o'clock position and zero the gauge.

Use a pry bar to raise the rear of the crankshaft to its upper limit. Record the value as (d) in the concentricity work sheet. This is the vertical bearing clearance adjustment and will **always** be positive.

Use the concentricity work sheet (see the accompanying illustration) to determine the total vertical and horizontal values.

The total horizontal is equal to the 9-o'clock reading (a) minus the 3-o'clock reading (c).

The total vertical is equal to the 12-o'clock reading (b) plus the bearing clearance (d).

Example:

- 6:00 o'clock = reference = 0
- 9:00 o'clock = [a] = 0.004
- 12:00 o'clock = [b] = 0.003
- 3:00 o'clock = [c] = -0.002.

Using the worksheet and the numbers from the example, the total horizontal value equals 0.006, and the total vertical value equals 0.005.

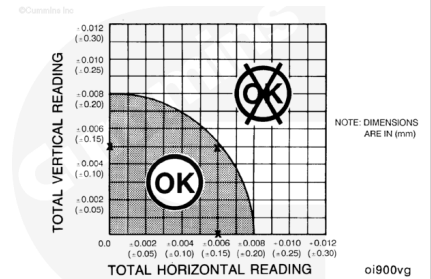
Concentricity Worksheet	
9 o'clock	a = 0.004
3 o'clock	c = -0.002
Total Horizontal	a - c = .006
12 o'clock	b = .003
Bearing Clearance	d = .002
Total Vertical	b + d = .005

Mark the total horizontal value on the horizontal side of the chart and the total vertical on the vertical side of the chart.

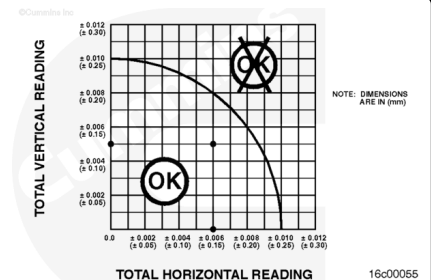
Use a straight edge to find the intersection point of the total horizontal and total vertical values. The intersection point **must** fall within the shaded area for the flywheel housing concentricity to be within specification.

Using the total horizontal and total vertical values from the previous example, the intersection point falls within the shaded area. Therefore, the flywheel housing concentricity is within specification.

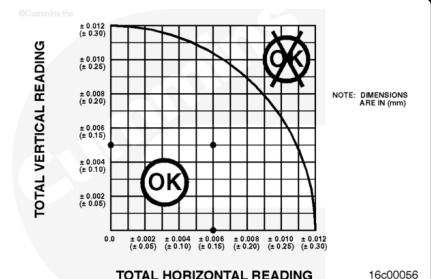
Use the accompanying chart for SAE 1, 2, and 3 flywheel housings.



Use the accompanying chart for SAE 0 and ½ flywheel housings.



Use the accompanying chart for SAE 00 flywheel housings.



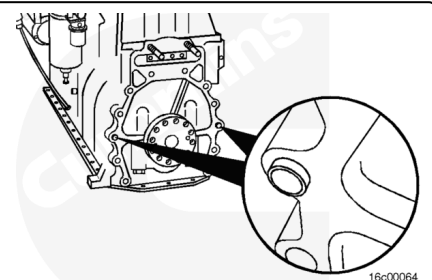
If the intersection point falls outside the shaded area, the ring dowels **must** be removed and the housing repositioned.

The ring dowels are **not** required to maintain concentricity of the housing; the clamping force of the capscrews holds the housing in place.

After the ring dowels are discarded, install the flywheel housing on the engine.

To position the housing, tighten the capscrews enough to hold the flywheel housing in place, but loose enough to enable small movement when struck lightly with a mallet.

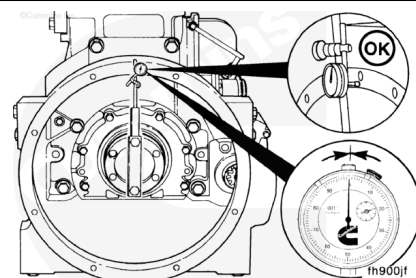
Check the concentricity again. When concentricity is within specification, tighten the capscrews to the specified value.



Face Alignment

⚠ CAUTION ⚠

The dial indicator tip must not enter the capscrew holes, or the gauge will be damaged.

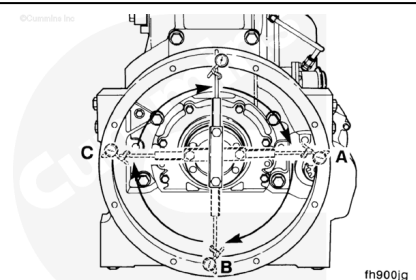


Install a dial indicator as illustrated.

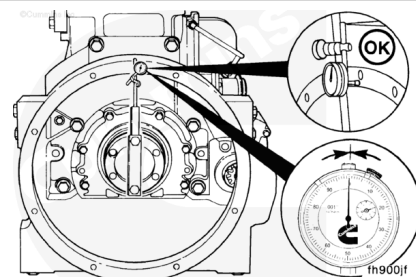
The extension bar for the indicator **must** be rigid for an accurate reading. It **must not** sag. Position the indicator at the 12-o'clock position. Adjust the dial until the needle points to zero.

Slowly rotate the crankshaft. Record the readings at the 3-o'clock, 6-o'clock, and 9-o'clock positions.

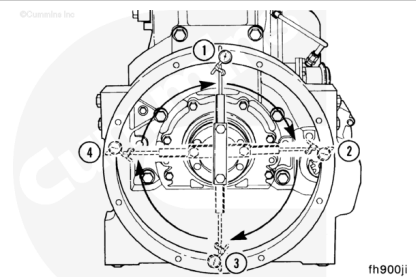
The crankshaft **must** be pushed toward the front of the engine to remove the crankshaft end clearance each time a position is measured.



Continue to rotate the crankshaft until the indicator is at the 12-o'clock position. Check the indicator to make sure that the needle points to zero. If it does **not**, the readings will be incorrect.



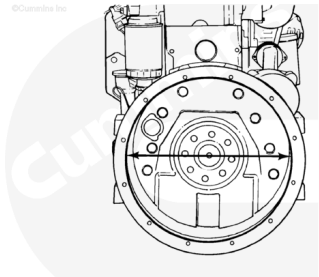
Determine the total indicator runout.



Total Indicator Runout (Example)

Position	Runout
12-o'clock	0.00 mm [0.000 in]
3-o'clock	+0.08 mm [+0.003 in]
6-o'clock	-0.05 mm [-0.002 in]
9-o'clock	+0.08 mm [+0.003 in]
Total Indicator Runout =	0.13 mm [0.005 in]

The maximum allowable total indicator reading is determined by the diameter of the housing bore. If the flywheel housing is out of specifications, replace the housing.

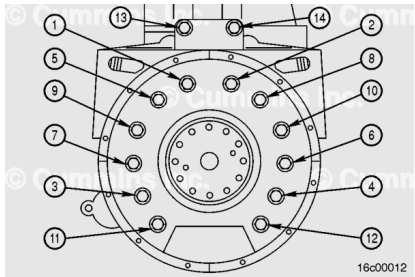


SAE Number	Bore Diameter	Total Indicator Runout Maximum
00	787.40 to 787.65 mm [30.88 to 31.00 in]	0.30 mm [0.012 in]
0	647.70 to 647.95 mm [25.50 to 25.51 in]	0.25 mm [0.010 in]
½	584.2 to 584.4 mm [23.000 to 23.008 in]	0.25 mm [0.010 in]
1	511.18 to 511.30 mm [20.125 to 20.130 in]	0.20 mm [0.008 in]
2	447.68 to 447.80 mm [17.625 to 17.630 in]	0.20 mm [0.008 in]
3	409.58 to 409.70 mm [16.125 to 16.130 in]	0.20 mm [0.008 in]

Install the remaining capscrews. Tighten all capscrews in two steps using the sequence shown.

Torque Value:

- 1. 100 n•m [74 ft-lb]
- 2. 197 n•m [145 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the rear engine mounts on the flywheel housing. Refer to Procedure 016-010 in Section 16. (</qs3/pubsys2/xml/en/procedures/00/00-016-010.html>)
- Install the starting motor capscrews, starter, and spacer. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/377/377-013-020-tr.html>)
- Install the rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (</qs3/pubsys2/xml/en/procedures/377/377-001-024.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-025.html>)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/377/377-007-037.html>)
- Install the flywheel. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/377/377-016-005.html>)
- Install the transmission, clutch, and all related components. See the equipment manufacturer service information.
- Connect the batteries and air starter. See the equipment manufacturer service information.
- Operate the engine to normal operating temperature and check for leaks and proper operation.